

EX NO:8
21.03.2025

DATE AND TIME FUNCTION

AIM:

To perform SQL date and time functions.

Table:

```
SQL>create table team(emp_id number(3), name varchar2(15), joining date,  
dept varchar2(10));
```

```
SQL> select * from team;
```

EMP_ID	NAME	JOINING	DEPT
101	Rahul	15-MAR-11	HR
102	Priya	22-JUL-13	Sales
103	Yokarajan	10-JAN-16	IT
104	Sneha	05-SEP-18	Admin
105	Vikram	30-NOV-19	HR
106	Neha	12-JUN-12	IT
107	Arjun	08-FEB-14	Sales
108	Pooja	25-APR-21	Admin

8 rows selected.

ADDING DATE:

```
SQL> select emp_id, name, joining, joining + interval '60' day as working  
from team;
```

EMP_ID	NAME	JOINING	WORKING
101	Rahul	15-MAR-11	14-MAY-11
102	Priya	22-JUL-13	20-SEP-13
103	Yokarajan	10-JAN-16	10-MAR-16
104	Sneha	05-SEP-18	04-NOV-18
105	Vikram	30-NOV-19	29-JAN-20

106 Neha	12-JUN-12 11-AUG-12
107 Arjun	08-FEB-14 09-APR-14
108 Pooja	25-APR-21 24-JUN-21

8 rows selected.

CURRENT DAY:

SQL> select name, sysdate as current_date from team;

NAME	CURRENT_D
-----	-----
Rahul	20-MAR-25
Priya	20-MAR-25
Yokarajan	20-MAR-25
Sneha	20-MAR-25
Vikram	20-MAR-25
Neha	20-MAR-25
Arjun	20-MAR-25
Pooja	20-MAR-25

8 rows selected.

TO FIND THE EXPERIENCE OF THE EMPLOYEE:

SQL> select name, floor(months_between(sysdate, joining)/12) as experience from team;

NAME	EXPERIENCE
-----	-----
Rahul	14
Priya	11
Yokarajan	9
Sneha	6
Vikram	5
Neha	12
Arjun	11
Pooja	3

EXTRACT:

```
SQL> select name, extract(year from joining) as joining_year from team;
```

NAME	JOINING_YEAR
Rahul	2011
Priya	2013
Yokarajan	2016
Sneha	2018
Vikram	2019
Neha	2012
Arjun	2014
Pooja	2021

8 rows selected.

TO FIND THE TOTAL NO OF DAYS USING TRUNC:

```
SQL> select name, trunc(sysdate) - trunc(joining) as total_days from team;
```

NAME	TOTAL_DAYS
Rahul	5119
Priya	4259
Yokarajan	3357
Sneha	2388
Vikram	1937
Neha	4664
Arjun	4058
Pooja	1425

8 rows selected.

TO FIND THE EMPLOYEE WHO HAS EXPERIENCE MORE THAN 10 YEARS:

```
SQL> select name from team where extract(year from sysdate) - extract(year from joining) > 10;
```

NAME

Rahul

Priya

Neha

Arjun

EMPLOYEES HIRED ON SAME DAY:

```
SQL> select t1.emp_id, t1.name, t1.joining, t2.emp_id, t2.name, t2.joining
from team t1 join team t2 on t1.joining = t2.joining and t1.emp_id >
t2.emp_id;
```

EMP_ID	NAME	JOINING	EMP_ID	NAME	JOINING
-----	-----	-----	-----	-----	-----
109	Ram	10-JAN-16	103	Yokarajan	10-JAN-16

EMPLOYEES HIRED ON ALTERNATE DAYS:

```
SQL> select t1.emp_id, t1.name, t1.joining, t2.emp_id, t2.name, t2.joining
from team t1 join team t2 on abs(t1.joining - t2.joining) = 1 and
t1.emp_id < t2.emp_id;
```

EMP_ID	NAME	JOINING	EMP_ID	NAME	JOINING
-----	-----	-----	-----	-----	-----
104	Sneha	05-SEP-18	110	Viji	06-SEP-18

TO FIND ON WHICH DAY OF THE YEAR THE EMPLOYEE IS HIRED:

```
SQL> select emp_id, name, to_char(joining, 'DDD') as hireday from team;
```

EMP_ID	NAME	HIR
-----	-----	---
101	Rahul	074
102	Priya	203
103	Yokarajan	010
104	Sneha	248
105	Vikram	334

106 Neha	164
107 Arjun	039
108 Pooja	115
109 Ram	010
110 Viji	249

10 rows selected.

TABLE:

```
SQL> create table events(event_id int, event_name varchar2(20), start_time
timestamp);
```

Table created.

TO VIEW THE TABLE

```
SQL> select event_id, event_name, to_char(start_time, 'HH24:MI:SS') as
start_time from events;
```

EVENT_ID	EVENT_NAME	START_TI
101	Seminar	09:00:00
102	Workshop	10:00:00
103	Coding Contest	11:10:00
104	Paper Presentation	12:55:00
105	Project Presentation	01:00:00

ADDING END TIME:

```
SQL> select event_id, event_name, to_char(start_time, 'HH24:MI:SS') as
start_time, to_char(start_time + interval '2' hour, 'HH24:MI:SS') as
end_time from events;
```

EVENT_ID	EVENT_NAME	START_TI	END_TIME
101	Seminar	09:00:00	11:00:00
102	Workshop	10:00:00	12:00:00
103	Coding Contest	11:10:00	13:10:00

104 Paper Presentation 12:55:00 14:55:00

105 Project Presentation 01:00:00 03:00:00

MARK SPLIT UP:

CONTENTS	MARKS ALLOTTED	MARKS OBTAINED
Aim, Algorithm, SQL, PL/SQL	30	
Execution and Result	20	
Viva	10	
Total	60	

RESULT:

The sql order by date and time operations are performed.